

# Unit 5 :: Introduction to Probability

☞ PHY0400404: Mathematical Physics ☞

Dr. Parag Bhattacharya  
Department of Physics  
Rangapara College (Autonomous)



# Syllabus

## Unit 5: Introduction to Probability

Independent random variables, probability distribution functions; binomial, Gaussian and Poisson distributions with examples; mean and variance.

# 1 Random variables

## **2 Probability distribution functions**

### **2.1 Binomial distribution**

### **2.2 Gaussian distribution**

### **2.3 Poisson distribution**

### **3 Statistics of a distribution**

#### **3.1 Mean of a distribution**

#### **3.2 Variance of a distribution**